

Supplemental Online Content

Sung H, Wiese D, Jatoi I, Jemal A. State variation in racial and ethnic disparities in incidence of triple-negative breast cancer among US women. *JAMA Oncol*. Published online March 2, 2023. doi:10.1001/jamaoncol.2022.7835

eTable1. Case number, age-standardized incidence rates, between racial and ethnic group disparities in incidence rates of triple-negative breast cancer by state in the United States, 2015-2019

eTable2. Within-racial and ethnic group disparities in incidence rates of triple-negative breast cancer by state in the United States, 2015-2019

eFigure 1. State-specific incidence rates of triple-negative breast cancer by race and ethnicity in the United States

eFigure 2. Pairwise correlation in state-specific incidence rates of triple-negative breast cancer among racial and ethnic populations in the United States, 2015-2019

This supplemental material has been provided by the authors to give readers additional information about their work.

eTable1. Case number, age-standardized incidence rates, between racial and ethnic group disparities in incidence rates of triple-negative breast cancer by state in the United States, 2015-2019

	All races and ethnicities combined		Non-Hispanic White (Reference)		Non-Hispanic American Indian/Alaska Native			Non-Hispanic Asian or Pacific Islander			Non-Hispanic Black			Hispanic (All Races)		
	Cases, N	Incidence rate per 100,000 women (SE)	Count	Incidence rate per 100,000 women (SE)	Cases, N	Incidence rate per 100,000 women (SE)	Incidence rate ratio (95% CI) (ref: White)	Cases, N	Incidence rate per 100,000 women (SE)	Incidence rate ratio (95% CI) (ref: White)	Cases, N	Incidence rate per 100,000 women (SE)	Incidence rate ratio (95% CI) (ref: White)	Cases, N	Incidence rate per 100,000 women (SE)	Incidence rate ratio (95% CI) (ref: White)
Total	133,579	13.9 (0.0)	86,195	12.9 (0.0)	768	11.2 (0.4)	0.86 (0.80 to 0.93)	4,969	9 (0.1)	0.70 (0.68 to 0.72)	28,710	25.2 (0.2)	1.95 (1.93 to 1.98)	12,937	11.1 (0.1)	0.86 (0.84 to 0.88)
Alabama	2,462	16.3 (0.3)	1,438	13.1 (0.4)	-	-	-	-	-	-	986	26.3 (0.9)	2 (1.84 to 2.18)	-	-	-
Alaska	225	12.4 (0.9)	157	12.4 (1)	41	16 (2.6)	1.29 (0.88 to 1.85)	-	-	-	-	-	-	-	-	-
Arizona	2,367	11.4 (0.2)	1,693	11.9 (0.3)	41	6.4 (1)	0.53 (0.38 to 0.73)	54	7.3 (1)	0.61 (0.46 to 0.81)	129	18.1 (1.6)	1.52 (1.26 to 1.83)	450	10.5 (0.5)	0.88 (0.79 to 0.98)
Arkansas	1,399	15.4 (0.4)	1,000	13.5 (0.5)	-	-	-	-	-	-	340	27 (1.5)	2 (1.75 to 2.28)	32	10.2 (2.1)	0.75 (0.48 to 1.12)
California	13,005	11.8 (0.1)	6,284	11.9 (0.2)	71	11.5 (1.4)	0.97 (0.75 to 1.24)	1,735	9 (0.2)	0.76 (0.72 to 0.8)	1,551	22.9 (0.6)	1.93 (1.82 to 2.05)	3,364	10.7 (0.2)	0.9 (0.86 to 0.94)
Colorado	1,981	12.7 (0.3)	1,531	12.6 (0.3)	-	-	-	48	8.5 (1.2)	0.67 (0.49 to 0.9)	96	17.4 (1.8)	1.38 (1.1 to 1.7)	291	12.5 (0.7)	0.99 (0.86 to 1.12)
Connecticut	1,631	14.3 (0.4)	1,170	13.5 (0.4)	-	-	-	37	8.6 (1.4)	0.63 (0.44 to 0.89)	260	24.4 (1.5)	1.81 (1.57 to 2.08)	162	14.1 (1.2)	1.04 (0.87 to 1.23)
Delaware	538	17.8 (0.8)	307	13.8 (0.9)	-	-	-	-	-	-	196	32 (2.4)	2.32 (1.9 to 2.81)	-	-	-
District of Columbia	371	20.5 (1.1)	86	14 (1.6)	-	-	-	-	-	-	263	25.5 (1.6)	1.82 (1.4 to 2.39)	-	-	-
Florida	8,866	13 (0.1)	5,241	12.1 (0.2)	-	-	-	140	7 (0.6)	0.58 (0.48 to 0.69)	1,933	22.1 (0.5)	1.83 (1.73 to 1.93)	1,541	10.7 (0.3)	0.88 (0.83 to 0.94)
Georgia	4,642	15.4 (0.2)	2,257	12.3 (0.3)	-	-	-	106	9.5 (1)	0.77 (0.62 to 0.95)	2,103	23.3 (0.5)	1.89 (1.78 to 2.01)	174	11.4 (1)	0.93 (0.78 to 1.1)
Hawaii	494	10.9 (0.5)	123	12.9 (1.3)	-	-	-	317	9.9 (0.6)	0.77 (0.61 to 0.99)	-	-	-	37	18.3 (3.2)	1.42 (0.92 to 2.13)
Idaho	621	13.1 (0.5)	558	13 (0.6)	-	-	-	-	-	-	-	-	-	44	13.9 (2.3)	1.06 (0.74 to 1.48)
Illinois	5,939	15.6 (0.2)	3,721	14.1 (0.2)	-	-	-	222	10.7 (0.7)	0.76 (0.66 to 0.87)	1,515	28.3 (0.7)	2.01 (1.89 to 2.14)	472	11.4 (0.5)	0.81 (0.73 to 0.89)
Indiana	2,958	14.9 (0.3)	2,384	13.9 (0.3)	-	-	-	-	-	-	456	27.9 (1.3)	2.01 (1.81 to 2.23)	86	11.9 (1.4)	0.86 (0.67 to 1.08)
Iowa	1,394	15.1 (0.4)	1,303	15.2 (0.4)	-	-	-	-	-	-	53	25.2 (3.6)	1.66 (1.22 to 2.2)	-	-	-
Kansas	1,194	14.4 (0.4)	993	14 (0.5)	-	-	-	-	-	-	117	27.9 (2.6)	1.98 (1.61 to 2.42)	55	9.4 (1.3)	0.67 (0.5 to 0.89)
Kentucky	2,013	15 (0.3)	1,732	14.3 (0.4)	-	-	-	-	-	-	249	26.6 (1.7)	1.86 (1.62 to 2.13)	-	-	-
Louisiana	2,642	19.3 (0.4)	1,279	14.2 (0.4)	-	-	-	-	-	-	1,278	31.2 (0.9)	2.2 (2.02 to 2.38)	60	13.5 (1.8)	0.95 (0.72 to 1.23)
Maine	527	11.5 (0.5)	510	11.4 (0.5)	-	-	-	-	-	-	-	-	-	-	-	-
Maryland	3,040	16.4 (0.3)	1,358	12.8 (0.4)	-	-	-	114	9.3 (0.9)	0.72 (0.59 to 0.88)	1,448	26 (0.7)	2.03 (1.88 to 2.2)	112	11.6 (1.2)	0.91 (0.73 to 1.11)
Massachusetts	2,808	13.2 (0.3)	2,179	12.6 (0.3)	-	-	-	103	9 (0.9)	0.72 (0.58 to 0.88)	285	21.4 (1.3)	1.7 (1.49 to 1.93)	240	15.2 (1)	1.21 (1.04 to 1.39)
Michigan	4,203	13.5 (0.2)	3,145	12.5 (0.2)	-	-	-	67	8.4 (1)	0.67 (0.52 to 0.86)	893	22.3 (0.8)	1.79 (1.65 to 1.93)	80	9.1 (1)	0.73 (0.57 to 0.91)
Minnesota	2,057	12.8 (0.3)	1,816	12.6 (0.3)	-	-	-	55	9.5 (1.3)	0.75 (0.56 to 0.99)	115	17.4 (1.7)	1.38 (1.12 to 1.68)	53	11.5 (1.8)	0.91 (0.66 to 1.24)
Mississippi	1,763	19.8 (0.5)	845	14.8 (0.5)	-	-	-	-	-	-	896	29.2 (1)	1.98 (1.79 to 2.18)	-	-	-
Missouri	2,948	15.7 (0.3)	2,244	14 (0.3)	-	-	-	31	9.1 (1.7)	0.65 (0.44 to 0.93)	627	31.9 (1.3)	2.28 (2.08 to 2.5)	42	10.2 (1.7)	0.73 (0.51 to 1)
Montana	391	12.1 (0.7)	358	12.1 (0.7)	-	-	-	-	-	-	-	-	-	-	-	-
Nebraska	690	12.9 (0.5)	593	12.6 (0.5)	-	-	-	-	-	-	46	23.2 (3.5)	1.85 (1.33 to 2.51)	35	13.7 (2.7)	1.09 (0.71 to 1.59)
New Hampshire	541	12.4 (0.6)	518	12.6 (0.6)	-	-	-	-	-	-	-	-	-	-	-	-
New Jersey	3,809	13.7 (0.2)	2,200	12.4 (0.3)	-	-	-	226	9.1 (0.6)	0.74 (0.64 to 0.85)	892	25.3 (0.9)	2.04 (1.88 to 2.22)	487	12 (0.6)	0.97 (0.87 to 1.07)
New Mexico	702	11.3 (0.4)	340	10.4 (0.6)	37	7.8 (1.3)	0.75 (0.51 to 1.07)	-	-	-	-	-	-	298	12.2 (0.7)	1.17 (0.98 to 1.39)
New York	8,544	14.1 (0.2)	4,726	12.7 (0.2)	-	-	-	520	10.5 (0.5)	0.82 (0.75 to 0.9)	2,089	23 (0.5)	1.81 (1.71 to 1.91)	1,194	13.2 (0.4)	1.04 (0.97 to 1.1)
North Carolina	5,360	17 (0.2)	3,246	14.4 (0.3)	34	9.8 (1.7)	0.68 (0.46 to 0.96)	71	9 (1.1)	0.63 (0.48 to 0.8)	1,851	28 (0.7)	1.94 (1.83 to 2.06)	158	11 (1)	0.77 (0.63 to 0.92)
North Dakota	239	11.8 (0.8)	212	11.3 (0.8)	-	-	-	-	-	-	-	-	-	-	-	-
Ohio	5,425	14.9 (0.2)	4,291	14 (0.2)	-	-	-	61	8.7 (1.1)	0.62 (0.47 to 0.8)	1,007	24.6 (0.8)	1.75 (1.63 to 1.88)	61	8 (1.1)	0.57 (0.43 to 0.74)
Oklahoma	1,682	14.7 (0.4)	1,228	13.8 (0.4)	165	17.9 (1.4)	1.3 (1.09 to 1.54)	-	-	-	202	26.7 (1.9)	1.93 (1.65 to 2.26)	66	11.9 (1.6)	0.86 (0.65 to 1.13)

Oregon	1,439	11.2 (0.3)	1,273	11.5 (0.3)	-	-	-	34	5.7 (1)	0.5 (0.34 to 0.7)	36	19.7 (3.4)	1.71 (1.18 to 2.39)	71	8.2 (1.1)	0.72 (0.54 to 0.93)
Pennsylvania	5,685	13.8 (0.2)	4,433	13 (0.2)	-	-	-	77	6.4 (0.8)	0.5 (0.39 to 0.63)	961	24.3 (0.8)	1.87 (1.74 to 2.02)	211	12.4 (0.9)	0.96 (0.83 to 1.1)
Rhode Island	409	11.9 (0.6)	332	11.6 (0.7)	-	-	-	-	-	-	31	18.7 (3.4)	1.61 (1.07 to 2.35)	35	11.1 (2)	0.96 (0.64 to 1.38)
South Carolina	2,722	17.2 (0.3)	1,546	13.9 (0.4)	-	-	-	-	-	-	1,108	27.4 (0.9)	1.96 (1.81 to 2.13)	46	9.6 (1.5)	0.69 (0.49 to 0.94)
South Dakota	272	11 (0.7)	254	11.5 (0.8)	-	-	-	-	-	-	-	-	-	-	-	-
Tennessee	3,116	15.1 (0.3)	2,195	13.1 (0.3)	-	-	-	-	-	-	841	26.9 (1)	2.06 (1.89 to 2.24)	55	10.6 (1.6)	0.81 (0.59 to 1.08)
Texas	9,688	13 (0.1)	4,838	12.2 (0.2)	43	16.1 (2.5)	1.32 (0.94 to 1.79)	322	8.6 (0.5)	0.7 (0.62 to 0.79)	2,125	23.9 (0.5)	1.95 (1.85 to 2.05)	2,360	10.6 (0.2)	0.87 (0.82 to 0.91)
Utah	643	9.2 (0.4)	556	9.2 (0.4)	-	-	-	-	-	-	-	-	-	68	9.8 (1.3)	1.06 (0.79 to 1.38)
Vermont	225	11.1 (0.8)	221	11.2 (0.8)	-	-	-	-	-	-	-	-	-	-	-	-
Virginia	3,873	15.3 (0.3)	2,327	13.4 (0.3)	-	-	-	130	7.8 (0.7)	0.58 (0.48 to 0.7)	1,278	26.8 (0.8)	2.01 (1.87 to 2.16)	130	8.9 (0.8)	0.66 (0.54 to 0.8)
Washington	2,559	11.9 (0.2)	2,075	12.1 (0.3)	45	15.4 (2.4)	1.28 (0.92 to 1.74)	185	8.7 (0.7)	0.72 (0.62 to 0.85)	115	18.2 (1.8)	1.51 (1.23 to 1.83)	139	10.1 (0.9)	0.84 (0.69 to 1.01)
West Virginia	886	14.8 (0.5)	841	14.8 (0.5)	-	-	-	-	-	-	38	22 (3.7)	1.48 (1.02 to 2.08)	-	-	-
Wisconsin	2,401	13.9 (0.3)	2,065	13.4 (0.3)	-	-	-	-	-	-	235	28.6 (1.9)	2.13 (1.84 to 2.45)	62	11.4 (1.6)	0.85 (0.63 to 1.11)
Wyoming	190	11.3 (0.9)	173	11.3 (0.9)	-	-	-	-	-	-	-	-	-	-	-	-

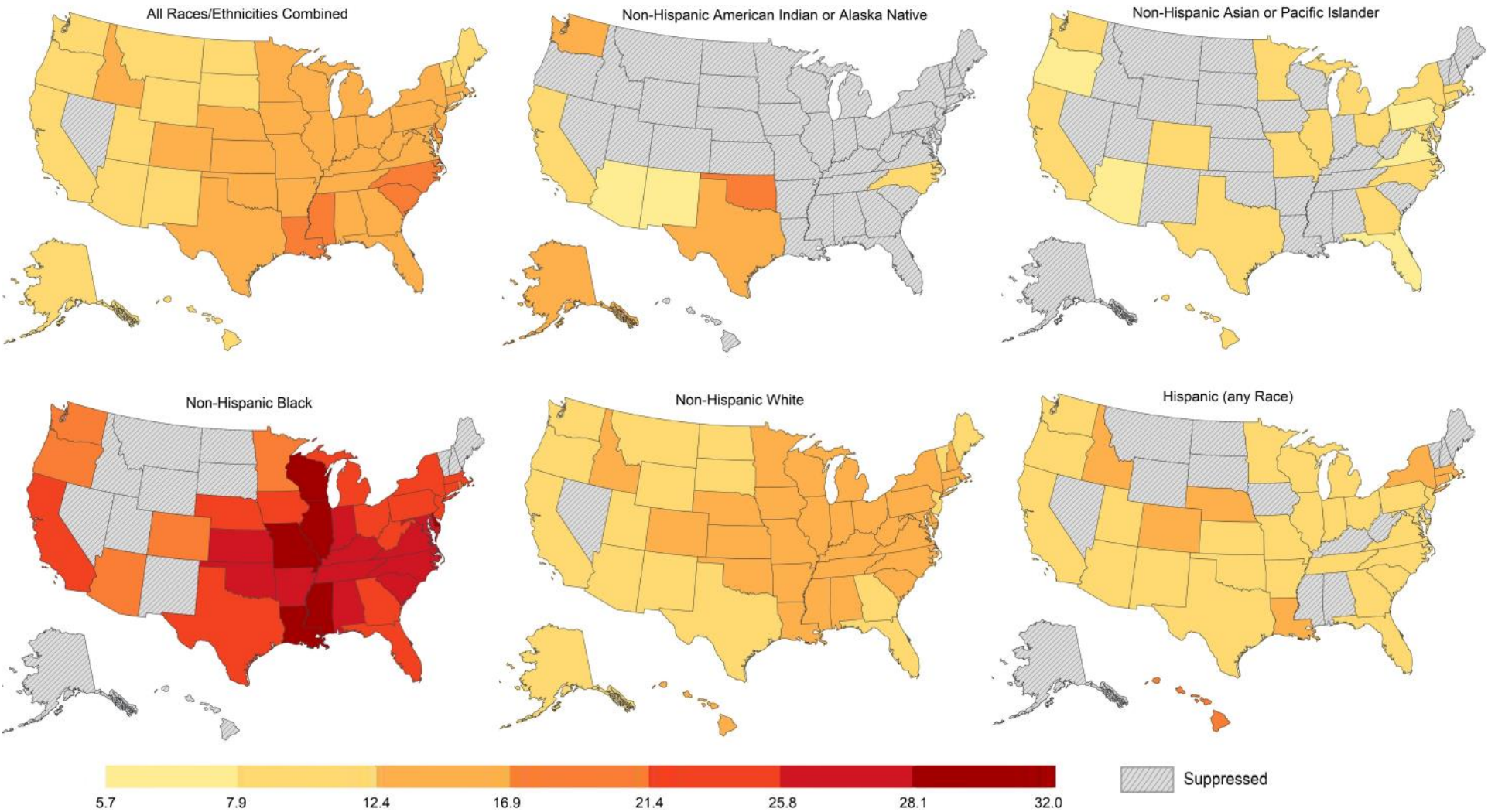
Abbreviations: CI, confidence interval; SE, standard error

For each racial and ethnic group, Nevada (due to reported data quality concerns) and states with less than <30 cases were excluded from the analysis and not shown.

eTable2. Within-racial and ethnic group disparities in incidence rates of triple-negative breast cancer by state in the United States, 2015-2019						
	All races and ethnicities combined	Non-Hispanic American Indian or Alaska Native	Non-Hispanic Asian or Pacific Islander	Non-Hispanic Black	Non-Hispanic White	Hispanic (All Races)
	Incidence rate ratio (95% CI)	Incidence rate ratio (95% CI)	Incidence rate ratio (95% CI)	Incidence rate ratio (95% CI)	Incidence rate ratio (95% CI)	Incidence rate ratio (95% CI)
Total	1 (reference)	1 (reference)	1 (reference)	1 (reference)	1 (reference)	1 (reference)
Alabama	1.17 (1.12 to 1.22)	-	-	1.05 (0.98 to 1.12)	1.02 (0.96 to 1.08)	-
Alaska	0.89 (0.78 to 1.02)	1.43 (1.01 to 1.97)	-	-	0.96 (0.81 to 1.14)	-
Arizona	0.82 (0.79 to 0.86)	0.57 (0.4 to 0.78)	0.81 (0.61 to 1.07)	0.72 (0.6 to 0.86)	0.92 (0.87 to 0.97)	0.95 (0.86 to 1.04)
Arkansas	1.1 (1.05 to 1.17)	-	-	1.07 (0.96 to 1.2)	1.05 (0.98 to 1.12)	0.92 (0.59 to 1.36)
California	0.85 (0.83 to 0.86)	1.03 (0.78 to 1.33)	1 (0.95 to 1.06)	0.91 (0.86 to 0.96)	0.92 (0.89 to 0.95)	0.97 (0.93 to 1)
Colorado	0.91 (0.87 to 0.95)	-	0.95 (0.69 to 1.27)	0.69 (0.56 to 0.85)	0.98 (0.93 to 1.03)	1.12 (0.99 to 1.27)
Connecticut	1.03 (0.98 to 1.08)	-	0.96 (0.67 to 1.33)	0.97 (0.85 to 1.1)	1.05 (0.99 to 1.12)	1.27 (1.07 to 1.49)
Delaware	1.28 (1.16 to 1.4)	-	-	1.27 (1.09 to 1.47)	1.07 (0.94 to 1.21)	-
District of Columbia	1.47 (1.32 to 1.64)	-	-	1.01 (0.89 to 1.15)	1.09 (0.86 to 1.36)	-
Florida	0.93 (0.91 to 0.95)	-	0.78 (0.65 to 0.92)	0.88 (0.84 to 0.92)	0.94 (0.91 to 0.97)	0.97 (0.92 to 1.02)
Georgia	1.11 (1.08 to 1.14)	-	1.06 (0.86 to 1.3)	0.92 (0.88 to 0.97)	0.96 (0.91 to 1)	1.03 (0.87 to 1.22)
Hawaii	0.78 (0.71 to 0.86)	-	1.11 (0.98 to 1.25)	-	1 (0.81 to 1.23)	1.66 (1.14 to 2.32)
Idaho	0.94 (0.87 to 1.02)	-	-	-	1.01 (0.93 to 1.11)	1.25 (0.88 to 1.72)
Illinois	1.12 (1.09 to 1.15)	-	1.19 (1.03 to 1.36)	1.13 (1.07 to 1.19)	1.09 (1.06 to 1.13)	1.03 (0.93 to 1.13)
Indiana	1.07 (1.03 to 1.11)	-	-	1.11 (1.01 to 1.22)	1.08 (1.03 to 1.12)	1.07 (0.84 to 1.34)
Iowa	1.08 (1.02 to 1.14)	-	-	1 (0.74 to 1.32)	1.18 (1.11 to 1.25)	-
Kansas	1.04 (0.97 to 1.1)	-	-	1.11 (0.91 to 1.33)	1.09 (1.02 to 1.16)	0.85 (0.63 to 1.12)
Kentucky	1.08 (1.03 to 1.13)	-	-	1.06 (0.92 to 1.2)	1.11 (1.05 to 1.17)	-
Louisiana	1.39 (1.33 to 1.44)	-	-	1.24 (1.17 to 1.31)	1.1 (1.04 to 1.17)	1.22 (0.93 to 1.57)
Maine	0.83 (0.75 to 0.91)	-	-	-	0.89 (0.8 to 0.97)	-
Maryland	1.18 (1.13 to 1.22)	-	1.03 (0.85 to 1.25)	1.03 (0.98 to 1.09)	0.99 (0.94 to 1.05)	1.05 (0.85 to 1.28)
Massachusetts	0.95 (0.91 to 0.99)	-	1.01 (0.82 to 1.23)	0.85 (0.75 to 0.96)	0.98 (0.94 to 1.03)	1.38 (1.2 to 1.57)
Michigan	0.97 (0.94 to 1)	-	0.93 (0.72 to 1.19)	0.89 (0.83 to 0.95)	0.97 (0.93 to 1.01)	0.82 (0.64 to 1.03)
Minnesota	0.92 (0.88 to 0.96)	-	1.05 (0.78 to 1.39)	0.69 (0.56 to 0.84)	0.98 (0.93 to 1.03)	1.04 (0.75 to 1.4)
Mississippi	1.43 (1.36 to 1.5)	-	-	1.16 (1.08 to 1.24)	1.15 (1.07 to 1.23)	-
Missouri	1.13 (1.09 to 1.18)	-	1.01 (0.68 to 1.45)	1.27 (1.17 to 1.38)	1.09 (1.04 to 1.14)	0.92 (0.65 to 1.26)
Montana	0.87 (0.78 to 0.96)	-	-	-	0.94 (0.84 to 1.05)	-
Nebraska	0.93 (0.86 to 1)	-	-	0.92 (0.67 to 1.24)	0.98 (0.89 to 1.06)	1.24 (0.81 to 1.79)
New Hampshire	0.89 (0.82 to 0.98)	-	-	-	0.98 (0.89 to 1.07)	-
New Jersey	0.98 (0.95 to 1.02)	-	1.02 (0.88 to 1.16)	1.01 (0.94 to 1.08)	0.96 (0.92 to 1.01)	1.08 (0.99 to 1.19)
New Mexico	0.81 (0.75 to 0.87)	0.7 (0.48 to 0.98)	-	-	0.81 (0.71 to 0.91)	1.1 (0.98 to 1.24)
New York	1.02 (0.99 to 1.04)	-	1.17 (1.06 to 1.28)	0.91 (0.87 to 0.96)	0.99 (0.96 to 1.02)	1.19 (1.12 to 1.26)
North Carolina	1.22 (1.19 to 1.26)	0.87 (0.59 to 1.25)	1 (0.77 to 1.28)	1.11 (1.06 to 1.17)	1.12 (1.08 to 1.16)	1 (0.83 to 1.19)
North Dakota	0.85 (0.74 to 0.97)	-	-	-	0.88 (0.76 to 1.01)	-
Ohio	1.07 (1.04 to 1.1)	-	0.96 (0.73 to 1.25)	0.98 (0.92 to 1.04)	1.09 (1.05 to 1.12)	0.72 (0.55 to 0.93)
Oklahoma	1.06 (1.01 to 1.11)	1.6 (1.34 to 1.91)	-	1.06 (0.91 to 1.22)	1.07 (1.01 to 1.14)	1.08 (0.81 to 1.39)
Oregon	0.81 (0.76 to 0.85)	-	0.64 (0.43 to 0.9)	0.78 (0.54 to 1.09)	0.89 (0.84 to 0.95)	0.74 (0.57 to 0.96)
Pennsylvania	0.99 (0.97 to 1.02)	-	0.72 (0.56 to 0.9)	0.97 (0.9 to 1.03)	1.01 (0.97 to 1.04)	1.12 (0.97 to 1.29)
Rhode Island	0.86 (0.77 to 0.95)	-	-	0.74 (0.5 to 1.06)	0.9 (0.8 to 1.01)	1 (0.69 to 1.41)
South Carolina	1.23 (1.19 to 1.28)	-	-	1.09 (1.02 to 1.16)	1.08 (1.02 to 1.14)	0.86 (0.62 to 1.17)

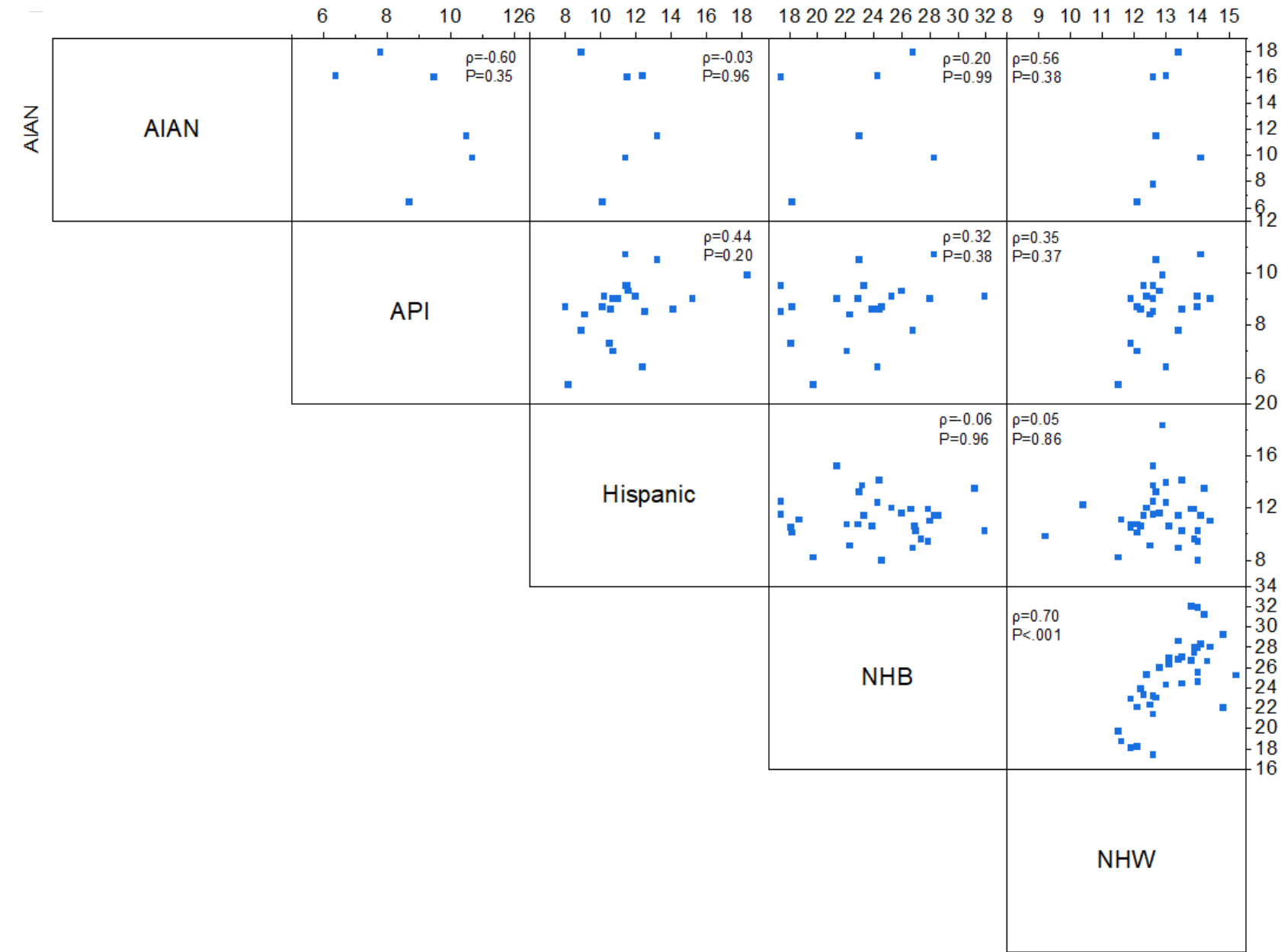
South Dakota	0.79 (0.69 to 0.89)	-	-	-	0.89 (0.78 to 1.02)	-
Tennessee	1.08 (1.04 to 1.12)	-	-	1.07 (1 to 1.15)	1.02 (0.97 to 1.06)	0.96 (0.7 to 1.27)
Texas	0.94 (0.92 to 0.96)	1.44 (1.02 to 1.98)	0.96 (0.85 to 1.08)	0.95 (0.91 to 0.99)	0.95 (0.92 to 0.98)	0.96 (0.91 to 1)
Utah	0.66 (0.61 to 0.72)	-	-	-	0.72 (0.66 to 0.78)	0.88 (0.67 to 1.13)
Vermont	0.8 (0.69 to 0.92)	-	-	-	0.87 (0.75 to 1.01)	-
Virginia	1.1 (1.06 to 1.14)	-	0.86 (0.72 to 1.03)	1.07 (1.01 to 1.13)	1.04 (0.99 to 1.08)	0.8 (0.66 to 0.96)
Washington	0.85 (0.82 to 0.89)	1.38 (0.99 to 1.89)	0.97 (0.83 to 1.13)	0.72 (0.59 to 0.88)	0.94 (0.89 to 0.98)	0.92 (0.76 to 1.1)
West Virginia	1.07 (0.99 to 1.14)	-	-	0.87 (0.61 to 1.22)	1.15 (1.07 to 1.24)	-
Wisconsin	1 (0.96 to 1.04)	-	-	1.14 (0.99 to 1.3)	1.04 (0.99 to 1.09)	1.03 (0.76 to 1.34)
Wyoming	0.81 (0.69 to 0.94)	-	-	-	0.88 (0.74 to 1.03)	-
Abbreviations: CI, confidence interval						
For each racial and ethnic group, Nevada (due to reported data quality concerns) and states with less than <30 cases were excluded from the analysis and not shown.						

eFigure 1. State-specific incidence rates of triple-negative breast cancer by race and ethnicity in the United States



NOTE States with less than 30 cases were suppressed and excluded from subsequent analyses. Nevada was excluded regardless of case number due to data quality concerns. Case number and age-standardized incidence rate (95% confidence interval) in each state stratified by race and ethnicity can be found in eTable1. Age-standardized incidence rates were calculated using the 2000 U.S. standard population. Maps were generated using QGIS 3.18.3.

eFigure 2. Pairwise correlation in state-specific incidence rates of triple-negative breast cancer among racial and ethnic populations in the United States, 2015-2019



Abbreviations: AIAN, non-Hispanic American Indian or Alaska Native; API, non-Hispanic Asian or Pacific Islander; Black, non-Hispanic Black; White, non-Hispanic White.

NOTE Spearman's Rho (ρ) was calculated to test correlations in state-specific rates between populations using SAS 9.4. P-values were adjusted for False Discovery Rate using Benjamini-Hochberg procedure.